

EDSON, AB, Canada

WMO: 711670

Lat: 53.5805N

Lon: 116.4535W

Elev: 3041

StdP: 13.15

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.3	-18.3	-33.5	0.9	-24.7	-26.0	1.4	-17.4	21.4	38.0	18.0	35.7	4.1	280	0.650

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.0	80.7	60.9	77.4	59.3	74.4	58.0	63.9	76.1	61.8	73.2	59.9	70.7	7.0	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.4	84.6	68.4	57.4	78.7	65.8	55.6	73.7	63.7	30.8	76.5	29.1	73.3	27.7	70.7	70.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.7	14.7	12.4	DB	-34.0	87.0	6.8	2.7	-38.9	88.9	-42.9	90.5	-46.7	92.0	-51.7	93.9	(4)
(5)			WB	-34.0	67.5	6.5	2.2	-38.7	69.1	-42.4	70.3	-46.1	71.6	-50.7	73.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.5	13.3	17.8	25.1	37.7	47.8	55.1	60.0	57.8	49.3	37.5	22.8	13.1
	DBStd	19.73	15.92	13.49	13.12	8.28	7.47	5.36	5.38	5.67	7.19	8.06	11.69	13.74
	HDD50	5784	1139	901	772	372	130	13	2	7	97	391	815	1145
	HDD65	10410	1604	1321	1237	818	532	301	170	230	471	851	1265	1610
	CDD50	874	0	0	0	4	62	165	313	248	77	5	0	0
	CDD65	25	0	0	0	0	0	3	16	6	0	0	0	0
	CDH74	746	0	0	0	2	50	97	339	199	56	3	0	0
	CDH80	120	0	0	0	0	6	13	65	31	5	0	0	0
Wind	WSAvg	5.0	4.5	4.7	5.2	5.7	6.0	5.6	5.3	4.7	4.8	4.7	4.2	4.6
Precipitation	PrecAvg	23.5	1.1	0.7	0.9	0.9	2.4	4.5	4.5	3.3	2.6	0.8	0.8	0.8
	PrecMax	32.5	3.2	2.0	2.0	2.4	7.1	10.0	9.0	7.6	6.0	1.8	1.8	1.9
	PrecMin	16.7	0.1	0.0	0.2	0.1	0.8	1.7	1.5	0.5	0.7	0.2	0.2	0.1
	PrecStd	5.1	0.9	0.4	0.5	0.6	1.4	2.0	2.3	2.0	1.5	0.4	0.6	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.5	50.7	56.9	70.3	79.9	81.7	86.2	83.7	80.2	69.7	53.5	47.4
		MCWB	38.5	38.1	42.4	49.3	54.3	59.5	62.9	63.4	58.8	51.9	42.3	37.3
	2%	DB	43.5	44.8	50.9	63.3	73.6	76.6	81.3	79.0	74.7	61.3	45.9	41.3
		MCWB	35.2	35.3	38.6	46.4	52.8	57.2	62.1	61.3	57.0	47.3	37.7	33.7
	5%	DB	38.8	40.8	46.9	58.3	69.4	73.0	77.8	75.6	70.0	55.8	41.3	36.2
		MCWB	32.5	32.8	36.2	43.4	51.3	55.8	60.6	60.2	55.3	44.5	34.9	30.7
	10%	DB	33.9	36.5	43.0	53.5	64.9	69.2	74.2	72.2	64.6	51.1	37.0	31.8
		MCWB	30.1	30.4	33.9	40.7	49.3	54.3	59.1	58.8	53.1	42.0	32.3	28.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.9	39.1	43.0	50.4	57.3	61.9	66.7	66.3	61.5	53.3	42.7	38.0
		MCDB	47.3	48.8	56.0	67.9	72.5	74.4	78.9	78.1	74.2	67.5	50.7	45.9
	2%	WB	35.9	35.9	39.4	47.3	54.6	59.8	64.4	63.9	58.4	48.9	38.7	34.4
		MCDB	42.8	44.1	49.4	61.7	69.6	72.1	77.4	75.3	71.0	59.2	44.7	40.4
	5%	WB	33.1	33.2	36.8	44.4	52.5	57.6	62.3	61.8	56.2	45.6	35.6	31.0
		MCDB	37.8	40.0	45.6	56.2	66.3	68.6	74.2	72.2	67.7	54.2	40.4	35.7
	10%	WB	29.9	30.8	34.6	41.6	50.4	55.7	60.3	59.7	53.7	42.9	32.8	27.9
		MCDB	34.0	36.2	42.1	51.8	63.2	66.0	71.3	69.8	63.9	49.9	36.2	31.6
Mean Daily Temperature Range	5% DB	MDBR	21.4	24.2	22.5	22.9	24.1	22.5	23.0	23.5	24.0	21.0	18.2	19.7
		MCDBR	25.4	28.8	28.1	33.2	34.2	31.0	30.5	31.1	35.6	30.3	23.2	22.6
		MCWBR	18.4	20.2	17.5	18.5	16.6	14.7	14.1	15.6	19.1	18.1	16.0	17.1
	5% WB	MCDBR	24.3	27.4	26.4	30.6	30.1	25.9	26.9	27.5	31.9	28.0	21.9	22.2
		MCWBR	18.4	19.4	16.9	17.6	15.3	13.0	13.2	14.9	18.0	17.6	15.9	17.5
Clear-Sky Solar Irradiance	taub		0.249	0.258	0.282	0.324	0.339	0.354	0.352	0.356	0.315	0.281	0.255	0.237
	taud		2.487	2.561	2.531	2.430	2.431	2.400	2.458	2.411	2.538	2.609	2.536	2.439
	Ebn at Noon		245	280	292	289	289	284	283	274	275	266	241	226
	Edh at Noon		15	20	25	32	34	35	33	32	25	19	15	13
All-Sky Solar Radiation	RadAvg		282	572	986	1380	1621	1726	1776	1505	1074	620	306	213
	RadStd		16	40	67	125	136	147	130	106	115	61	23	16

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (27 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page